

Henry Francis Ring

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SUMMARY

Software engineer with 7+ years delivering cloud platforms, satellite and ground systems, hardware-in-the-loop automation, and embedded avionics. Combines full-stack development with systems integration, test automation, and field deployment across AWS, Linux, and real hardware.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Rust, C/C++, C#, Ada, SQL

Cloud & Web: AWS (Kinesis Video Streams, Lambda, ECS, AppSync, DynamoDB, CloudWatch, SSM, Secrets Manager, IAM, S3, Amplify), React, Next.js, Node.js, GraphQL, REST, WebSockets

Systems & DevOps: Linux, Docker, Ansible, Tauri, GStreamer, PostgreSQL, Azure, Git/CI-CD, HITL, SNMP/SCPI, RTOS/VxWorks

EXPERIENCE

Software Integration & Test Engineer, Amazon Leo (Contract) Nov 2024 - Present

- Served as the sole developer for a bespoke multi-site AWS video platform supporting secure-site monitoring, remote hardware inspections, and safety checks; owned GStreamer edge pipelines, Kinesis Video Streams, Lambda/AppSync services, and a React/TypeScript frontend on Amplify; replaced a ~\$30/camera/month vendor tool and VPN-based monitoring; built it to scale to dozens of cameras and enable future adoption across additional programs, with projected savings of thousands per month for one program alone.
- Designed keyless IAM/OIDC/SAML-to-STS credential federation that removed static credentials from field devices; built a Tauri/React/Rust cloud-infrastructure provisioning tool that codified AWS SSM managed-instance registration and host configuration across macOS, Windows, and Ubuntu, cutting cloud provisioning time by ~95%.
- Automated HITL rack commissioning and fleet operations across FPGA/FEP flashing, certificates, GGMA firmware, PDU/PSU/relay control, and Cisco Catalyst 9300 configuration with Python and Ansible.
- Executed test plans for local and remote site-activation campaigns covering FEP-to-GGMA test plans, rack/network/GNSS validation, satellite telemetry and commanding, and NTP drift/encryption stability; authored 30+ runbooks and closed 100+ work items across five programs.

Senior Software Engineer, Quadrata, Inc. Jul 2024 - Sep 2024

- Built and maintained Python APIs documented with OpenAPI/Swagger and PostgreSQL data services for decentralized identity and compliance credentials across DeFi/Web3 platforms; containerized environments with Docker Compose for repeatable delivery.
- Delivered responsive Next.js/React interfaces with Radix UI and Mantine, integrated front- and back-end systems, and strengthened quality through peer reviews and technical documentation.

Senior Software Engineer, BigBear.ai (Contract) Sep 2023 - Mar 2024

- Built React/Node/Python decision-support applications, REST APIs, and real-time WebSocket services; developed a federated search engine with machine-learning classification and Azure/IaC deployment workflows for scalable delivery.

Software Engineer II, Northrop Grumman Space Systems Jul 2022 - Mar 2023

- Developed build/integration tooling and spacecraft-simulator updates for NGGSTE, improving telemetry packet processing, networking reliability, and execution on hardware-in-the-loop satellite test systems.

Software Engineer, Raytheon Jan 2019 - May 2022

- Built an award-recognized C#/ .NET mission-file tool and nightly DevOps pipeline that reduced generation time from days to minutes; trained engineers and supported program-wide adoption.
- Developed and tested embedded C/C++/Ada/Python software, multithreaded built-in test (BIT), and power diagnostics on APG-79/APG-82 radar hardware across TCP/IP, MIL-STD-1553, Ethernet, fiber, RS-232, and ARINC interfaces under DO-178C processes.

Software Development Intern, Jet Propulsion Laboratory (JPL) Jan 2018 - May 2018

- Built a React/Node/MongoDB web app with UC Merced's Mechatronics, Embedded Systems and Automation (MESA) Lab to control a methane-sensing drone and visualize flight and mapping data.

EDUCATION

B.S., Computer Science & Engineering | University of California, Merced | 2018